

Highlights

- Collapse–recovery dynamics are simulated across ten 1000-year civilizational futures.
- Duty cycles (fraction of time a civilization remains active) range from 0.38 to 1.00.
- Resource depletion rate and recovery fraction are generally the most impactful resilience levers.
- An effective detectability duration reformulates the Drake Equation’s longevity term.
- Low duty cycles help explain the apparent absence of extraterrestrial signals.